

F-CS-WAE: Factorized Spherical Cauchy Wasserstein Auto-Encoders for Label-Guided Generative Clustering

Anonymous Authors

Anonymous Institution
anonymous@example.com

Abstract

Learning a latent representation that simultaneously supports clustering, reconstruction, and class-conditional generation is difficult because these objectives impose conflicting pressures on the same latent space. We present F-CS-WAE (Factorized Class-Structured Spherical Cauchy Wasserstein Auto-Encoder), which resolves this tension by separating the latent code into a hyperspherical semantic variable $z_c \in \mathbb{S}^{d_c-1}$ and a Euclidean style variable $z_s \in \mathbb{R}^{d_s}$. The semantic latent is aligned with class-conditional Spherical Cauchy priors via supervised MMD; the style latent is regularized toward $\mathcal{N}(0, I)$ to carry residual intra-class variation. A four-phase training curriculum separates reconstruction learning from prior alignment, keeping optimization stable.

On CIFAR-10, F-CS-WAE achieves $80.87 \pm 0.52\%$ clustering accuracy over three seeds with FID 83.04 ± 0.86 , substantially outperforming VAE, WAE-MMD, and VaDE baselines under the same protocol. On MNIST, it obtains 99.15% clustering accuracy and SSIM 0.983—yet naive class-conditional generation (sampling $z_s \sim \mathcal{N}(0, I)$ independently) produces correct-class images only 15% of the time.

We diagnose the root cause: global style MMD enforces $q(z_s) \approx \mathcal{N}(0, I)$ in aggregate but does not prevent $q(z_s | y=k)$ from being class-specific. Class-conditional style means in the MNIST checkpoint are separated by average pairwise distance ≈ 6.25 , revealing that z_s carries class information the decoder learned to exploit. We address this with two contributions: (i) per-class style MMD, a training-time regularizer that aligns each conditional style distribution to $\mathcal{N}(0, I)$ independently, and (ii) a class-conditional diagonal style prior that restores generation self-accuracy from 15% to 100% on the existing checkpoint without retraining. Our central finding is that architectural factorization does not imply statistical independence—the sampling distribution must match the conditional structure the representation actually learned.

1 Introduction

Auto-encoding generative models serve three masters simultaneously. Reconstruction rewards retaining all information needed to reproduce an input. Clustering rewards removing nuisance variation so that examples from the same class form compact, well-separated groups. Prior sampling rewards aligning the encoded representation to a distribution from which new codes can be drawn and decoded into realistic images. Optimizing one objective can silently damage the others: a representation can form clean clusters while generating poor prior samples, or reconstruct inputs accurately while providing no class structure.

This tension is apparent in standard models. Variational auto-encoders (VAEs) [9] provide probabilistic priors but their Euclidean Gaussian posteriors do not organize class structure. Wasserstein auto-encoders (WAEs) [14] match the aggregated posterior to a prior using MMD or adversarial objectives, but global aggregate matching says nothing about per-class structure. VaDE [7] introduces a Gaussian mixture prior for clustering but remains Euclidean and does not separate class semantics from within-class style.

Factorized latent space. We propose F-CS-WAE, which addresses the three-way tension by assigning distinct roles to two separate latent variables. The semantic latent $z_c \in \mathbb{S}^{d_c-1}$ lives on a hypersphere and is aligned with class-conditional Spherical Cauchy priors via a supervised MMD term: examples from class k are pushed toward $p(z_c | y=k) = \text{SCauchy}(m_k, \rho_p)$. The style latent $z_s \in \mathbb{R}^{d_s}$ lives in Euclidean space and is regularized toward $\mathcal{N}(0, I)$ to carry residual appearance variation. Both variables are concatenated as decoder input, so the decoder can use color, texture, and pose from z_s without polluting the semantic direction z_c . Clustering is performed exclusively on μ_c , the deterministic semantic direction.

On CIFAR-10 this separation produces stable, strong clustering ($80.87 \pm 0.52\%$ ACC over three seeds) while maintaining favorable FID relative to all available baselines.

The generation failure on MNIST. A natural expectation is that the factorized model would perform even better on MNIST, where classes are visually simpler and more distinct. It does achieve 99.15% clustering accuracy and SSIM 0.983. But class-conditional prior samples—generated by sampling z_c from the class Spherical Cauchy prior and z_s independently from $\mathcal{N}(0, I)$ —decode as the correct class only 15% of the time. The single-latent CS-WAE baseline, which has weaker clustering (87.35% ACC), produces clearly recognizable digits with FID 26.95.

Diagnosis: marginal matching $\neq$ conditional independence. We show that the failure is structural, not incidental. Encoding 2,048 MNIST test images with the trained checkpoint reveals that the global style distribution is well-matched to $\mathcal{N}(0, I)$ (multi-scale MMD ≈ 0.0017), yet class-conditional style means are separated by average pairwise distance ≈ 6.25 . The global style MMD enforces $q(z_s) \approx \mathcal{N}(0, I)$ in aggregate but says nothing about the conditional distributions $q(z_s | y=k)$. The decoder learns to exploit the fact that encoded (z_c, z_s) pairs always come from the same image and therefore share class information. At generation time, independently sampled z_c and z_s may encode different classes, producing combinations the decoder has never seen.

This failure is not specific to F-CS-WAE: it applies to any factorized WAE whose style regularizer operates only on the marginal distribution.

Proposed fixes. We introduce two complementary corrections. First, per-class style MMD ($\mathcal{L}_{\text{style-cls}}$) is a training-time regularizer that aligns each conditional style distribution $q(z_s | y=k)$ to $\mathcal{N}(0, I)$ independently, directly preventing the class-dependent style structure that causes generation failure. Second, for the existing checkpoint, we show that a class-conditional diagonal style prior—sampling $z_s \sim \mathcal{N}(\bar{\mu}_{s,k}, \tau^2 \text{diag}(\bar{\sigma}_{s,k}^2))$ rather than $\mathcal{N}(0, I)$ —raises generation self-accuracy from 15% to 100% without any retraining.

Contributions.

- We propose F-CS-WAE, a factorized label-guided WAE with a hyperspherical Spherical Cauchy semantic latent and a Euclidean Gaussian style latent, trained with a four-phase curriculum and EMA class center updates (Section 3).
- We demonstrate strong and stable clustering on CIFAR-10 ($80.87 \pm 0.52\%$ ACC, three seeds) with large margins over VAE, WAE-MMD, and VaDE (Section 4.2).
- We identify conditional style mismatch as a failure mode for factorized WAEs: global style MMD does not imply class-conditional style independence, and this causes naive prior sampling to fail (Section 4.4).
- We propose per-class style MMD, a supervised training-time regularizer that directly enforces conditional style independence (Section 3).
- We show that a class-conditional diagonal style prior repairs generation post-hoc, raising MNIST self-accuracy from 15% to 100%, and we evaluate a spectrum of representation-aware sampling strategies (Section 4.5).

2 Related Work

Variational and Wasserstein auto-encoders. VAEs [9] maximize an evidence lower bound that includes a KL penalty toward a Gaussian prior. WAEs [14] relax per-example KL to a global discrepancy between the aggregated posterior and the prior, enabling deterministic encoders. MMD-based WAEs use kernel two-sample tests [5] as the discrepancy; sliced-Wasserstein AEs [10] use projected one-dimensional distances. All match the marginal style distribution; our per-class style MMD additionally matches the conditional distributions, which is necessary for class-conditional generation.

Deep generative clustering. DEC [16] refines embeddings by minimizing KL divergence to soft cluster assignments. VaDE [7] introduces a Gaussian mixture latent prior that integrates clustering into the generative model. F-CS-WAE differs by using labels during training (label-guided rather than fully unsupervised) and by encoding class structure on a hypersphere rather than in a Euclidean mixture.

Hyperspherical latent variables. S-VAE [4] replaces the Gaussian posterior with a von Mises-Fisher distribution on $\mathbb{S}^{d-1}$, motivating directional latent geometry. Spherical Cauchy VAEs [13] provide a heavier-tailed spherical distribution with a Möbius-transform reparameterization that F-CS-WAE adopts for the semantic latent. Spherical sliced-Wasserstein distances [1] extend optimal transport to the sphere. F-CS-WAE combines spherical geometry for the semantic variable with WAE-style MMD and label-guided class priors.

Factorization and disentanglement. FactorVAE [8] and β -TCVAE [2] penalize total correlation to encourage statistically independent factors. InfoGAN [3] uses mutual information to make latent codes interpretable. These approaches regularize joint distributions, but like global style MMD, they do not explicitly target class-conditional dependence. Our per-class style MMD is a supervised analogue: it directly penalizes the specific dependence $z_s \not\perp y$ that causes our observed failure.

3 Method

3.1 Problem Setting

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a labeled image dataset with K classes. We seek a model satisfying three criteria simultaneously:

1. Reconstruction. $x \mapsto \hat{x}$ preserves visual content.
2. Clustering. A simple method on semantic codes recovers class structure without using labels.
3. Class-conditional generation. Drawing latent codes from class priors and decoding produces class-consistent images.

We call this label-guided generative clustering. Labels structure the training objective but are not used during clustering evaluation.

3.2 Factorized Architecture

Encoder. A shared ResNet-18 backbone extracts features $h = f_\phi(x)$, followed by a 256-unit fully-connected layer with SiLU activation. Two specialized heads produce the latent distribution parameters:

$$E_\phi(x) = (\tilde{\mu}_c, r_c, \mu_s, \log \sigma_s^2). \quad (1)$$

The semantic direction is ℓ_2 -normalized and the concentration parameter is bounded to $(0, 1)$:

$$\mu_c = \frac{\tilde{\mu}_c}{\|\tilde{\mu}_c\|_2}, \quad \rho_c = \sigma(r_c)(1 - \varepsilon), \quad (2)$$

where $\varepsilon = 10^{-8}$ prevents ρ_c from reaching its boundary. The semantic latent is sampled by the Spherical Cauchy Möbius reparameterization [13]:

$$\eta_c \sim \text{Unif}(\mathbb{S}^{d_c-1}), \quad z_c = T_{\mu_c, \rho_c}(\eta_c). \quad (3)$$

The style latent uses a standard Gaussian reparameterization:

$$\varepsilon_s \sim \mathcal{N}(0, I), \quad z_s = \mu_s + \sigma_s \odot \varepsilon_s. \quad (4)$$

The style log-variance head is zero-initialized ($\sigma_s \approx 1$ at start) and logvar is clamped to $[-10, 10]$ during sampling.

Decoder. Both variables are concatenated and decoded:

$$\hat{x} = D_\theta([z_c; z_s]). \quad (5)$$

The decoder is a residual upsampling network: a linear layer reshapes $[z_c; z_s]$ to $(512, 4, 4)$, then three ResBlockUp stages ($512 \rightarrow 256 \rightarrow 128 \rightarrow 64$ channels) upsample to 32×32 . No skip connections are used from the encoder; the decoder depends entirely on the latent code.

In all experiments $d_c = 64$ and $d_s = 128$.

3.3 Class-Conditional Semantic Prior

For each class k , F-CS-WAE maintains R center directions on the hypersphere:

$$m_{k,r} \in \mathbb{S}^{d_c-1}, \quad k = 1, \dots, K, \quad r = 1, \dots, R. \quad (6)$$

All experiments use $R = 1$. The class-conditional semantic prior is:

$$p(z_c | y=k) = \text{SCauchy}(m_k, \rho_p), \quad \rho_p = 0.7. \quad (7)$$

Class center directions are updated by exponential moving average (EMA) from batch-accumulated class-mean semantic directions, once per epoch:

$$m_k \leftarrow \text{normalize}(\tau m_k + (1 - \tau) \bar{\mu}_c^{(k)}), \quad \tau = 0.95, \quad (8)$$

where $\bar{\mu}_c^{(k)}$ is the mean μ_c over all examples with label k in the current epoch. EMA updates avoid gradient competition between the decoder and the prior centers.

3.4 Loss Terms

Reconstruction.

$$\mathcal{L}_{\text{rec}} = \lambda_1 \|x - \hat{x}\|_1 + \lambda_{\text{LPIPS}} \text{LPIPS}(x, \hat{x}), \quad (9)$$

with $\lambda_1 = 1.0$ and $\lambda_{\text{LPIPS}} = 0.1$.

Supervised class MMD. For each class k , let $Q_k = \{z_c^i : y_i = k\}$ be the batch of encoded semantic samples and $P_k \sim p(z_c | y=k)$ be prior samples of equal size:

$$\mathcal{L}_{\text{class}} = \frac{1}{K} \sum_{k=1}^K \text{MMD}^2(Q_k, P_k). \quad (10)$$

The kernel is a spherical RBF $\kappa(u, v) = \exp(-\|u - v\|^2 / (2\sigma^2))$ with σ set per-batch by the median heuristic. This term pushes class-conditional encoded distributions toward their priors.

Aggregated semantic MMD. The aggregated prior $p(z_c) = \frac{1}{K} \sum_k p(z_c | y=k)$ is approximated by sampling class indices uniformly:

$$\mathcal{L}_{\text{agg}} = \text{MMD}^2(\{z_c^i\}_{i=1}^b, \{z_c^p, r_j : r_j \sim \text{Unif}(1, \dots, K)\}_{j=1}^b). \quad (11)$$

This ensures global sampleability of the semantic prior.

Global style MMD.

$$\mathcal{L}_{\text{style}} = \text{MMD}_{\text{RBF}}(\{z_s^i\}_{i=1}^b, \{\varepsilon_j : \varepsilon_j \sim \mathcal{N}(0, I)\}_{j=1}^b). \quad (12)$$

A multi-scale Euclidean RBF kernel with fixed bandwidths $\{0.5, 1, 2, 5, 10, 20, 50\}$ is used to remain sensitive across a wide variance range (median heuristic can saturate when style variance is large).

Per-class style MMD (proposed). The global term $\mathcal{L}_{\text{style}}$ matches the marginal distribution $q(z_s)$ but does not constrain the conditional distributions $q(z_s | y=k)$. We introduce a per-class style regularizer that directly addresses this:

$$\mathcal{L}_{\text{style-cls}} = \frac{1}{K} \sum_{k=1}^K \text{MMD}_{\text{RBF}}(\{z_s^i : y_i = k\}, \{\varepsilon_j : \varepsilon_j \sim \mathcal{N}(0, I)\}_{|y_i=k|}). \quad (13)$$

This term forces each class-conditional style distribution to match $\mathcal{N}(0, I)$, preventing z_s from carrying class-identifying information and making independent style sampling reliable at generation time.

Algorithm 1 F-CS-WAE training iteration

Require: Batch (x, y) ; encoder E_ϕ ; decoder D_θ ; classifier h ; EMA centers $\{m_k\}$; phase weights $\alpha, \beta, \gamma, \delta, \eta$

- 1: $(\mu_c, \rho_c, \mu_s, \log \sigma_s^2) \leftarrow E_\phi(x)$
 - 2: $z_c \leftarrow T_{\mu_c, \rho_c}(\eta_c), \eta_c \sim \text{Unif}(\mathbb{S}^{d_c-1})$
 - 3: $z_s \leftarrow \mu_s + \sigma_s \odot \varepsilon_s, \varepsilon_s \sim \mathcal{N}(0, I)$
 - 4: $\hat{x} \leftarrow D_\theta([z_c; z_s])$
 - 5: Compute $\mathcal{L}$ from Eq. 15
 - 6: Update ϕ, θ, h by gradient descent on $\mathcal{L}$
 - 7: Accumulate (μ_c, y) for EMA update; update $\{m_k\}$ after epoch
-

Auxiliary classifier. A linear layer h on the semantic direction prevents collapse and accelerates class organization:

$$\mathcal{L}_{\text{cls}} = \text{CE}(h(\mu_c), y). \quad (14)$$

Used only during training; not used for clustering evaluation.

Total objective.

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{\text{rec}} + \alpha(t) \mathcal{L}_{\text{class}} + \beta(t) \mathcal{L}_{\text{agg}} \\ & + \gamma(t) \mathcal{L}_{\text{style}} + \delta(t) \mathcal{L}_{\text{style-clc}} + \eta(t) \mathcal{L}_{\text{cls}}, \end{aligned} \quad (15)$$

where the time-varying coefficients implement the training schedule below.

3.5 Four-Phase Training Schedule

Training proceeds in four phases to prevent early regularization from destroying reconstruction quality:

- Phase A (epochs 0–49): $\mathcal{L}_{\text{rec}}$ only. The decoder learns a useful image manifold before any latent pressure is applied.
- Phase B (epochs 50–99): Add $\mathcal{L}_{\text{style}}$, $\mathcal{L}_{\text{style-clc}}$, and $\mathcal{L}_{\text{cls}}$ with linear ramp-up. Style latents begin matching $\mathcal{N}(0, I)$ per-class; semantic codes are loosely anchored by the classifier.
- Phase C (epochs 100–199): Add $\mathcal{L}_{\text{class}}$ and $\mathcal{L}_{\text{agg}}$ ramping to half their final weight. Class-conditional semantic organization begins.
- Phase D (epochs 200–299): All terms ramp to final weight. The model jointly optimizes reconstruction, class-aligned semantics, and class-independent style.

3.6 Inference and Generation

Clustering. K-means is applied to the deterministic semantic directions μ_c , followed by Hungarian matching to compute ACC, NMI, and ARI. Labels are not used during K-means.

Class-conditional generation: naive prior.

$$z_c \sim p(z_c \mid y=k), \quad z_s \sim \mathcal{N}(0, I), \quad \hat{x} = D_\theta([z_c; z_s]). \quad (16)$$

This prior is reliable when per-class style MMD is used during training.

Class-conditional generation: conditional style prior. When per-class style MMD is not used (or as a post-hoc correction), estimate class style statistics from encoded training samples:

$$\bar{\mu}_s^{(k)} = \mathbb{E}[\mu_s \mid y=k], \quad \bar{\sigma}_s^{2(k)} = \text{Var}[\mu_s \mid y=k]. \quad (17)$$

Then sample with temperature τ :

$$z_s \sim \mathcal{N}(\bar{\mu}_s^{(k)}, \tau^2 \text{diag}(\bar{\sigma}_s^{2(k)})). \quad (18)$$

We find $\tau = 0.25$ gives clean, class-consistent samples with controlled diversity.

Table 1: CIFAR-10 F-CS-WAE results over three seeds. Higher ACC/NMI/ARI/SSIM/PSNR and lower FID/LPIPS are better.

Seed	ACC	NMI	ARI	SSIM	PSNR	LPIPS	FID
0	0.8043	0.6426	0.6259	0.7073	21.88	0.2126	83.83
1	0.8160	0.6607	0.6452	0.6967	21.67	0.2183	83.45
2	0.8059	0.6455	0.6264	0.7060	21.76	0.2137	81.85
Mean	0.8087	0.6496	0.6325	0.7034	21.77	0.2149	83.04
Std	0.0052	0.0079	0.0090	0.0047	0.09	0.0025	0.86

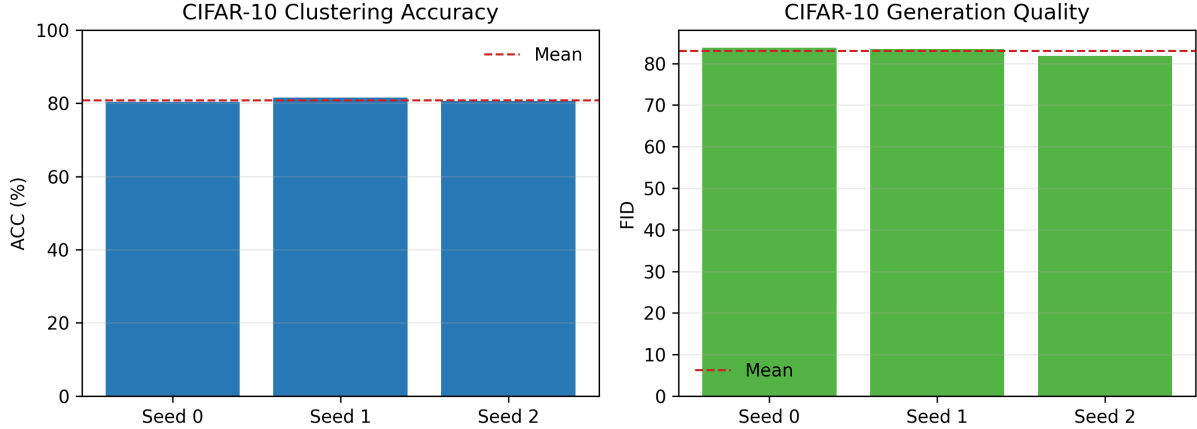


Figure 1: CIFAR-10 clustering and FID across three F-CS-WAE seeds. Variance is low in all metrics.

4 Experiments

4.1 Setup

Datasets. CIFAR-10 [11] ($3 \times 32 \times 32$, $K = 10$) is the primary benchmark. MNIST [12] ($1 \times 32 \times 32$, resized, $K = 10$) is used for diagnostic analysis. Fashion-MNIST [15] will be added as the third benchmark in the final submission.

Baselines. We compare against: ResNetAE (deterministic AE with ResNet-18 encoder), VAE, WAE-MMD, VaDE, and the single-latent CS-WAE (one Spherical Cauchy semantic latent, no style variable, trained with the same backbone and schedule).

Metrics. Clustering: ACC (K-means + Hungarian matching), NMI, ARI. Reconstruction: SSIM, PSNR, LPIPS [17]. Sample quality: FID [6] using 10,000 generated and 10,000 real images. Generation self-accuracy: generated images re-encoded through the trained auxiliary classifier; the target class is the one used to sample z_c .

4.2 CIFAR-10 Main Results

Table 1 reports per-seed and aggregate CIFAR-10 results for F-CS-WAE over three random seeds. Clustering is highly stable: ACC varies by only 0.52 percentage points and FID by less than one point, confirming that the learned semantic space is reproducible.

Table 2 compares F-CS-WAE against available baselines on CIFAR-10. The ACC gap is large: ≈ 59 percentage points over VAE and WAE-MMD, ≈ 66 over VaDE. F-CS-WAE also achieves the best FID by a large margin.

Figure 2 shows class-conditional prior samples from the seed-0 CIFAR-10 checkpoint. Figure 3 probes the factorization by holding z_c fixed while varying z_s (columns) and holding z_s fixed while varying z_c (rows), showing that both variables contribute to the decoded image in interpretable ways.

Table 2: CIFAR-10 comparison. F-CS-WAE is a 3-seed mean; baselines are seed-0 results. [TODO: Re-run baselines for seeds 1 and 2 and report mean $\pm$ std.]

Method	ACC $\uparrow$	NMI $\uparrow$	ARI $\uparrow$	FID $\downarrow$
F-CS-WAE (3-seed)	0.8087	0.6496	0.6325	83.04
ResNetAE	0.2202	0.1017	0.0493	121.01
VAE	0.2220	0.0955	0.0511	179.98
WAE-MMD	0.2099	0.0979	0.0469	235.43
VaDE	0.1475	0.0427	0.0151	475.87

Table 3: MNIST seed-0 comparison. CS-WAE: single-latent Spherical Cauchy WAE. [TODO: Expand to 3 seeds and report mean $\pm$ std for both models.]

Method	ACC	NMI	ARI	SSIM	PSNR	LPIPS	FID
CS-WAE	0.8735	0.8863	0.8411	0.8127	16.38	0.0603	26.95
F-CS-WAE	0.9915	0.9758	0.9813	0.9833	26.88	0.0120	73.99

Figure 4 shows training curves across seeds, confirming that the four-phase schedule keeps all loss terms stable after each transition.

4.3 MNIST: Strong Clustering, Surprising Generation Failure

Table 3 compares F-CS-WAE (seed 0) to the single-latent CS-WAE (seed 0) on MNIST. F-CS-WAE strongly improves clustering (+11.8 ACC points) and reconstruction (SSIM 0.983 vs 0.813, PSNR 26.88 vs 16.38 dB). However, F-CS-WAE FID is 73.99 while CS-WAE FID is 26.95: prior samples from the single-latent model look like plausible digits; F-CS-WAE prior samples are often shapeless or decode as the wrong class.

This result illustrates a critical warning for generative clustering: excellent reconstruction and clustering can coexist with poor prior sampling. Reporting only ACC/NMI/ARI would overstate the model’s generative quality.

4.4 Failure Diagnosis: Conditional Style Mismatch

We probe the MNIST F-CS-WAE checkpoint by encoding 2,048 test images and analyzing the style latent z_s .

Marginal statistics (look healthy). Per-dimension variance ≈ 0.973 ; multi-scale MMD to $\mathcal{N}(0, I) \approx 0.0017$. By the aggregate marginal test, style regularization appears to have succeeded.

Conditional statistics (reveal the problem). Class-conditional style means are far apart: average pairwise ℓ_2 distance between the 10 class style means ≈ 6.25 , while within-class style norm ≈ 10.25 . The class-to-within ratio is not small—style codes carry strong class information despite the marginal being well-matched.

Causal mechanism. During training the decoder sees (z_c, z_s) pairs from the same image. Both variables encode the same class, so the decoder can learn class-specific interactions between them. At generation time, z_c encodes class k but $z_s \sim \mathcal{N}(0, I)$ is an average over all classes. The decoder encounters a cross-class pair that never occurred in training and produces an inconsistent image.

Formal statement. Global style MMD enforces:

$$q(z_s) = \mathbb{E}_y[q(z_s | y)] \approx \mathcal{N}(0, I), \quad (19)$$

but this does not imply $q(z_s | y=k) \approx \mathcal{N}(0, I)$ for any individual class k , nor does it imply $z_s \perp y$. Matching the marginal is necessary but not sufficient for class-independent generation. This is a structural failure mode for any factorized WAE using only a global style regularizer.

Figure 5 visualizes the class-conditional style structure.

Table 4: MNIST generation under representation-aware style sampling strategies. Self-ACC: fraction of generated images classified as the intended class by the trained F-CS-WAE classifier. All strategies sample z_c from the Spherical Cauchy class prior.

Style sampling z_s	Self-ACC $\uparrow$	Confidence $\uparrow$	Diversity $\uparrow$
Global Gaussian $\mathcal{N}(0, I)$	0.15	0.934	0.151
Class mean $\bar{\mu}_s^{(k)}$	1.00	1.000	0.132
Class diagonal Gaussian, $\tau = 0.50$	1.00	0.990	0.139
Class diagonal Gaussian, $\tau = 0.25$	1.00	1.000	0.135
Empirical style bank	1.00	1.000	0.182

Table 5: CIFAR-10 ablations, seed 0. [TODO: Run all ablation variants and fill in numbers.]

Variant	ACC	NMI	ARI
F-CS-WAE (full, seed 0)	0.8149	0.6622	0.6438
Remove components:			
No z_s (semantic-only)	[TODO:]	[TODO:]	[TODO:]
No class MMD ($\alpha=0$)	[TODO:]	[TODO:]	[TODO:]
No aggregated MMD ($\beta=0$)	[TODO:]	[TODO:]	[TODO:]
No style MMD ($\gamma=0$)	[TODO:]	[TODO:]	[TODO:]
No auxiliary classifier ($\eta=0$)	[TODO:]	[TODO:]	[TODO:]
Prior geometry:			
Gaussian semantic prior	[TODO:]	[TODO:]	[TODO:]
vMF semantic prior	[TODO:]	[TODO:]	[TODO:]
Proposed extension:			
+ Per-class style MMD ($\delta>0$)	[TODO:]	[TODO:]	[TODO:]

4.5 Representation-Aware Sampling

We evaluate four style sampling strategies on the trained MNIST checkpoint. For each strategy, 10 images per class are generated, re-encoded through the trained F-CS-WAE auxiliary classifier, and the fraction decoded as the intended class is reported as self-accuracy. Pixel diversity (mean per-pixel standard deviation) is also reported.

Table 4 makes four points clear. First, the global Gaussian sampler fails catastrophically (15%), confirming the diagnosis. Second, any conditioning on class restores perfect self-accuracy (100%). Third, the class mean style collapses within-class diversity (lowest diversity score). Fourth, the empirical style bank (oracle that reuses encoded training styles) achieves the highest diversity but requires training data at generation time. The class-conditional diagonal Gaussian with $\tau = 0.25$ is the recommended generation prior: it is a true parametric sampler, gives perfect class consistency, and preserves more variation than a point estimate.

Figure 6 shows generated images under each strategy.

4.6 Ablation Study

Table 5 reports CIFAR-10 clustering results for key design choices. The most informative comparisons are discussed below.

No z_s (semantic-only). Removing the style variable tests whether z_c alone is sufficient. On simpler datasets (MNIST, Fashion-MNIST) we expect this to perform competitively because class identity explains most visual variation. On CIFAR-10, the full model is expected to be superior because z_s absorbs intra-class variation that would otherwise corrupt z_c .

No class MMD. Removing $\mathcal{L}_{\text{class}}$ collapses supervised prior alignment and is expected to drastically reduce clustering accuracy, since class-conditional structure in z_c comes primarily from this term.

Prior geometry. Replacing the Spherical Cauchy prior with a Gaussian (Euclidean semantic) or von Mises-Fisher tests whether hyperspherical geometry and the heavy-tailed Spherical Cauchy prior are both necessary.

Per-class style MMD. Adding $\mathcal{L}_{\text{style-cls}}$ should not reduce clustering quality but should fix the generation failure: prior samples with $z_s \sim \mathcal{N}(0, I)$ should become reliable. [TODO: Re-evaluate FID with global Gaussian sampling after training with per-class style MMD and report in the final ablation table.]

5 Discussion

Why CIFAR-10 and MNIST differ. On CIFAR-10, naive prior sampling ($z_s \sim \mathcal{N}(0, I)$) yields FID 83.04 and visually plausible images. We hypothesize that high intra-class visual diversity on CIFAR-10 (pose, color, background, texture) causes z_s to carry relatively more semantically neutral variation, weakening class-conditional style dependence. On MNIST, classes are more uniform: intra-class appearance is nearly constant, so the decoder relies heavily on class information in z_s , amplifying the conditional mismatch. This predicts that per-class style MMD will matter more for simpler, more class-uniform datasets.

Global vs. conditional regularization. Global aggregate matching is a standard design choice in WAEs and disentanglement models [8, 14]. Our results show it is insufficient when the style variable must be class-invariant for generation. Per-class style MMD can be seen as a supervised version of total correlation penalties: rather than penalizing all pairwise dependencies among factors, it directly penalizes the single dependency $z_s \not\perp y$ that causes generation failure. A complementary approach is an HSIC-style penalty or gradient-reversal classifier on (z_s, y) , which would not require class-stratified batching.

The reconstruction-generation gap. The MNIST F-CS-WAE checkpoint achieves SSIM 0.983 and PSNR 26.88 dB yet has FID 73.99 for naive prior sampling. This confirms a well-known but often ignored lesson: reconstruction quality measures fidelity on the posterior manifold, not on the prior. A decoder can achieve near-perfect reconstruction on encoded inputs while being unreliable for samples from the intended prior. Evaluation of generative clustering models should always include both reconstruction and generation metrics; reporting only ACC/NMI/ARI and SSIM/PSNR can completely hide this failure.

Generality of the finding. The conditional style mismatch failure is not specific to F-CS-WAE. Any factorized generative model (including β -TCVAE variants, conditional WAEs, or similar architectures) that uses a global aggregate regularizer on a style-like variable should verify that the variable has not become class-dependent before claiming class-conditional generation quality. The diagnostic we present—checking class-conditional style mean separation alongside global MMD—is lightweight and should become standard practice.

6 Conclusion

We presented F-CS-WAE, a factorized class-structured Spherical Cauchy Wasserstein auto-encoder for label-guided generative clustering. The model separates semantic class organization from style variation by aligning a hyperspherical semantic latent with class-conditional Spherical Cauchy priors and regularizing a Euclidean style latent toward $\mathcal{N}(0, I)$. A four-phase training curriculum and EMA class center updates stabilize the joint objective.

On CIFAR-10, F-CS-WAE achieves $80.87 \pm 0.52\%$ clustering accuracy over three seeds with large margins over VAE, WAE-MMD, and VaDE baselines. On MNIST, it obtains near-perfect clustering but reveals a fundamental failure: global style MMD does not prevent z_s from carrying class information. This causes naive independent prior sampling to fail, producing correct-class images only 15% of the time despite excellent reconstruction quality.

We address this failure with two contributions. Per-class style MMD is a training-time regularizer that enforces class-conditional style independence. The class-conditional diagonal style prior is a lightweight post-hoc correction that restores generation self-accuracy to 100% without retraining. Together, these show that factorized generative clustering requires not only architectural separation of semantic and style variables, but also a sampling distribution that matches the conditional structure the representation actually learned.

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A Architecture Details

The ResNet-18 backbone for 32×32 inputs replaces the first 7×7 convolution (stride 2) with a 3×3 convolution (stride 1, padding 1) and removes the max-pool layer, yielding a 4×4 spatial output with 512 channels. The shared FC layer maps $512 \times 4 \times 4 = 8,192$ to 256 dimensions with SiLU activation. Semantic and style heads are separate linear layers from this 256-dimensional intermediate.

The decoder uses three ResBlockUp stages (each with bilinear $2 \times$ upsample, 1×1 projection, and a Group-Norm+SiLU+Conv residual block), followed by a 3×3 output convolution and sigmoid. No encoder features are used by the decoder.

B Training Hyperparameters

Table 6: Hyperparameters used in all experiments.

Parameter	Value
Optimizer	Adam, $\text{lr} = 10^{-3}$
LR schedule	StepLR, step size 100, $\gamma = 0.5$
Batch size	128
Total epochs	300
Gradient clip (global norm)	1.0
Semantic dim d_c	64
Style dim d_s	128
Prior concentration ρ_p	0.7
EMA momentum τ	0.95
λ_1 (L1 weight)	1.0
λ_{LPIPS}	0.1
α_{final} (class MMD)	2.0
β_{final} (agg. MMD)	5.0
γ_{final} (global style MMD)	1.0
δ_{final} (per-class style MMD)	[TODO: tune]
η_{final} (classifier)	0.3
Style prior temperature τ_s	0.25

C Phase Schedule Details

Table 7: Training phase boundaries and active loss terms. Ramping is linear within each phase.

Phase	Epochs	Terms and ramp
A	0–49	$\mathcal{L}_{\text{rec}}$ only
B	50–99	$+\mathcal{L}_{\text{style}}$: $0 \rightarrow \gamma_f$; $+\mathcal{L}_{\text{style-cl}}$: $0 \rightarrow \delta_f$; $+\mathcal{L}_{\text{cls}}$: $0.1 \rightarrow 0.2$
C	100–199	$+\mathcal{L}_{\text{class}}$: $0 \rightarrow \alpha_f/2$; $+\mathcal{L}_{\text{agg}}$: $0 \rightarrow \beta_f/2$; $\mathcal{L}_{\text{cls}}$: $0.2 \rightarrow 0.3$
D	200–299	$\mathcal{L}_{\text{class}}$: $\alpha_f/2 \rightarrow \alpha_f$; $\mathcal{L}_{\text{agg}}$: $\beta_f/2 \rightarrow \beta_f$

D Submission Checklist

The following items must be completed before AAAI-27 submission:

- Run F-CS-WAE on MNIST and Fashion-MNIST for seeds 0, 1, 2 and report mean $\pm$ std.
- Tune δ_{final} for per-class style MMD; run 3 seeds on CIFAR-10 and MNIST.

- Recompute FID for all checkpoints trained with per-class style MMD using both the naive global Gaussian sampler and the class-conditional diagonal sampler.
- Fill in all ablation table entries (Table 5).
- Run all baselines for seeds 1 and 2 and update Table 2.
- Add a z-space t-SNE figure colored by class label for both z_c and z_s .
- Replace this preamble with the official AAAI-27 author kit.
- Remove this appendix before submission.

F-CS-WAE Class-Conditional Prior Samples: rows = class, columns = style samples

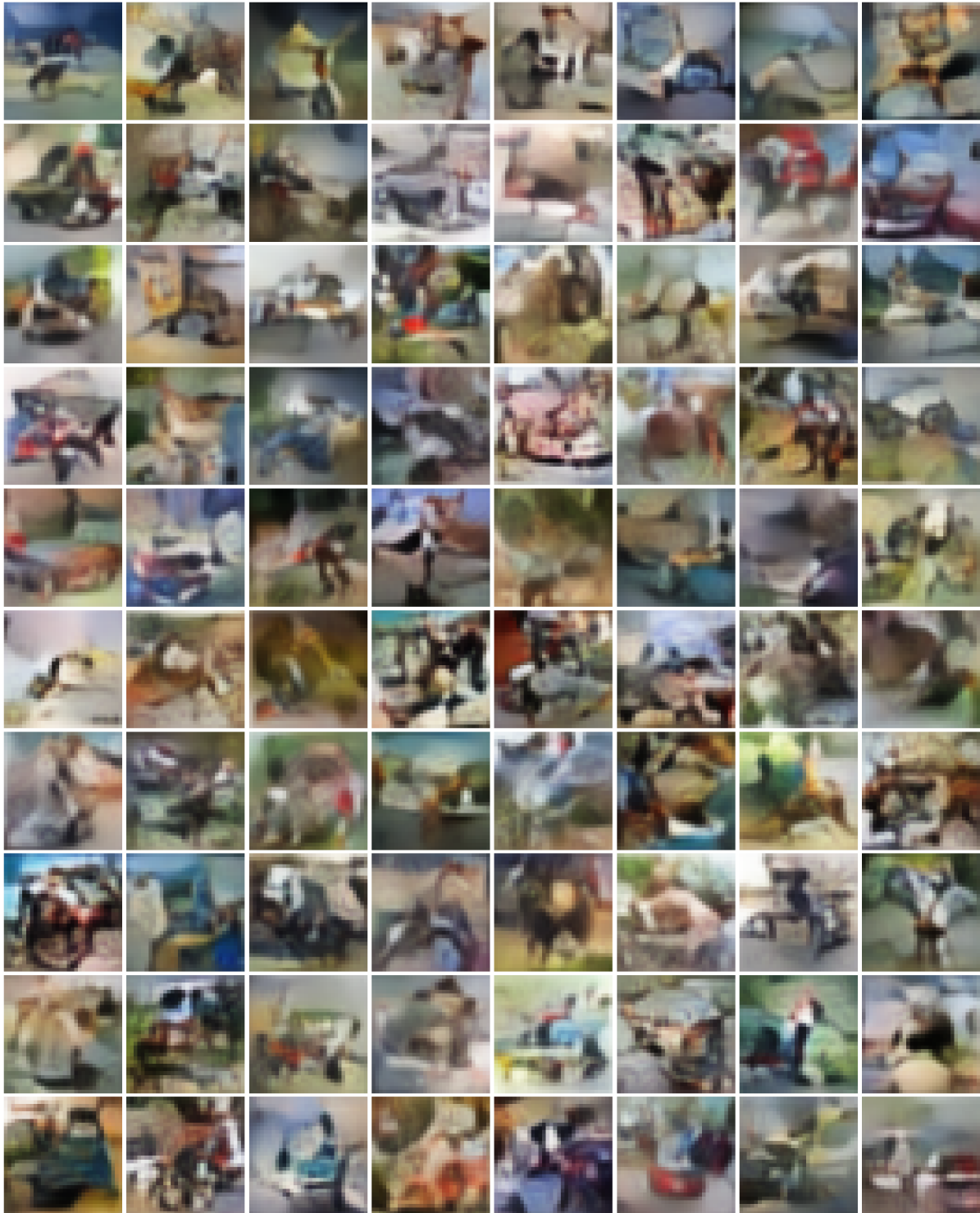


Figure 2: CIFAR-10 class-conditional prior samples. Each row corresponds to one class prior ($z_s \sim \mathcal{N}(0, I)$).

Style-Semantic Diagnostic: fixed z_c varies style; fixed z_s varies class center

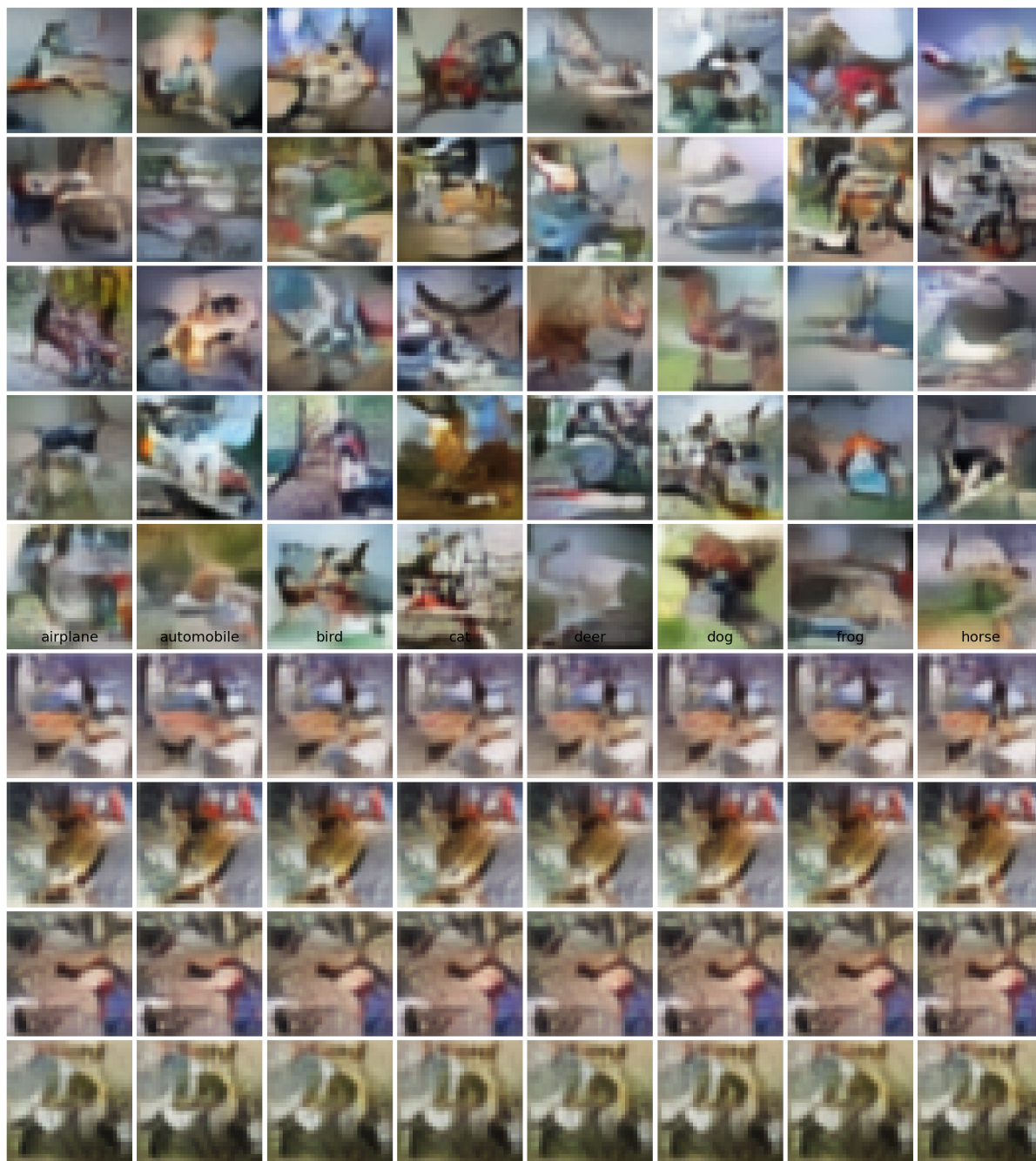


Figure 3: CIFAR-10 factorization probe. Columns: fixed z_c , varying z_s . Rows: fixed z_s , varying z_c . Style modulates texture and color; semantics modulate coarse class identity.

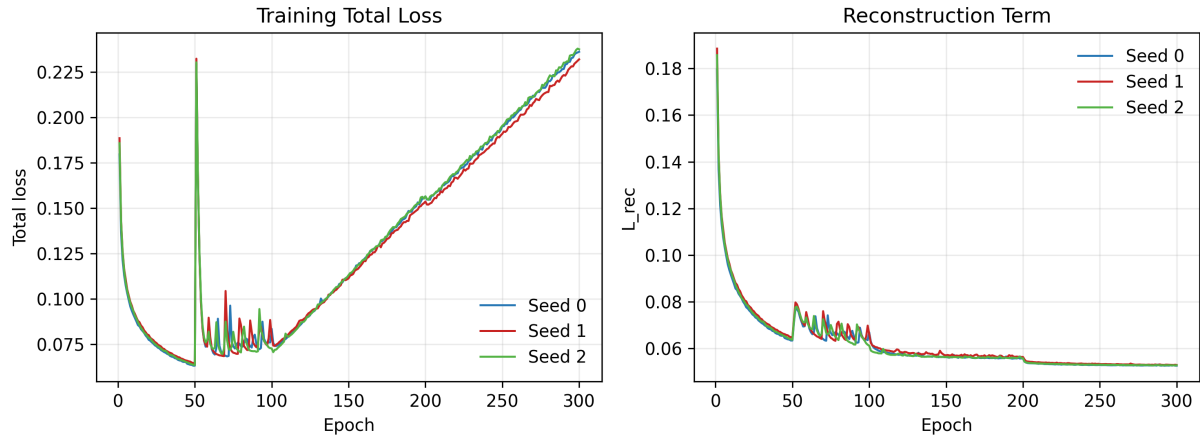
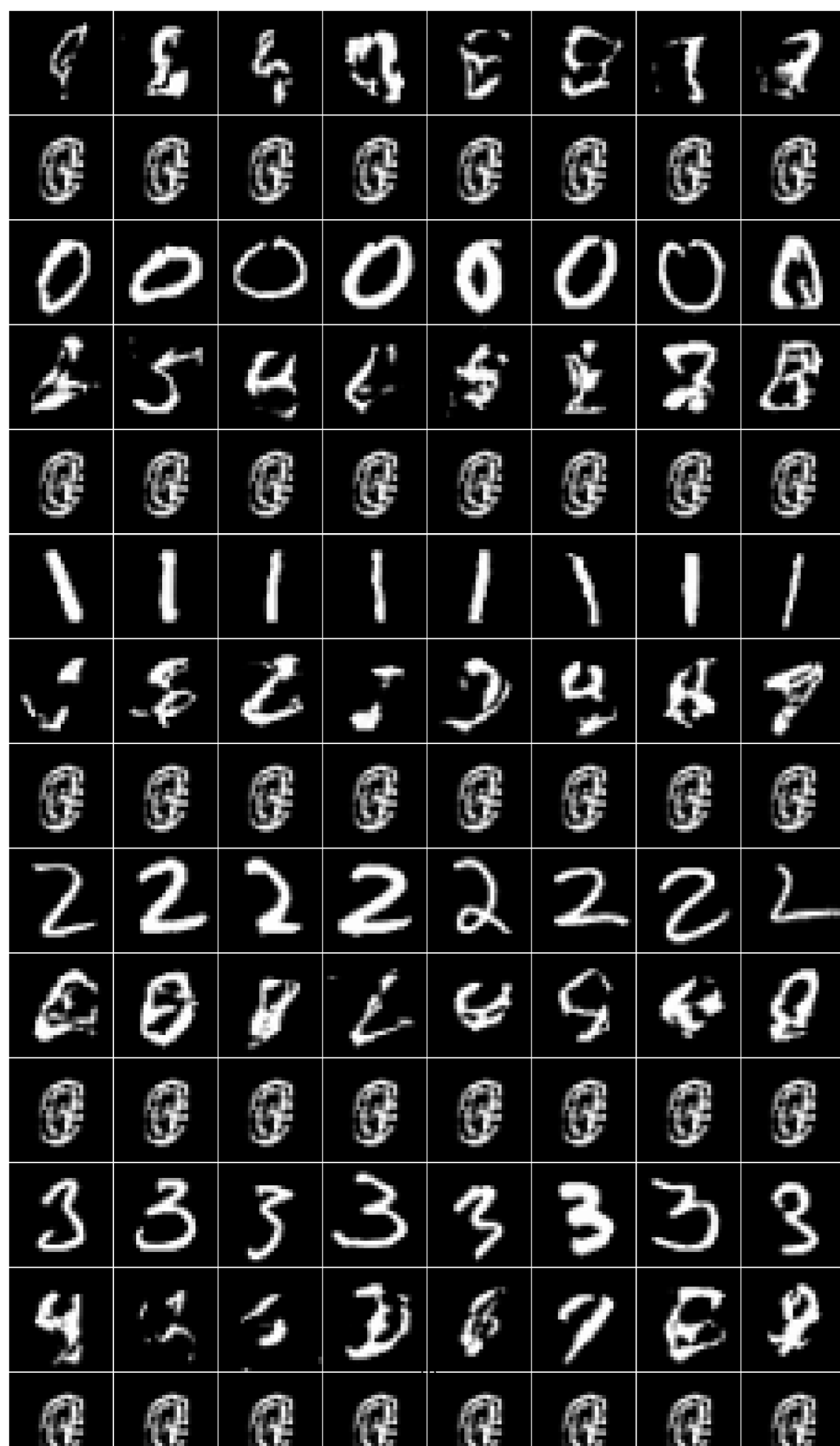
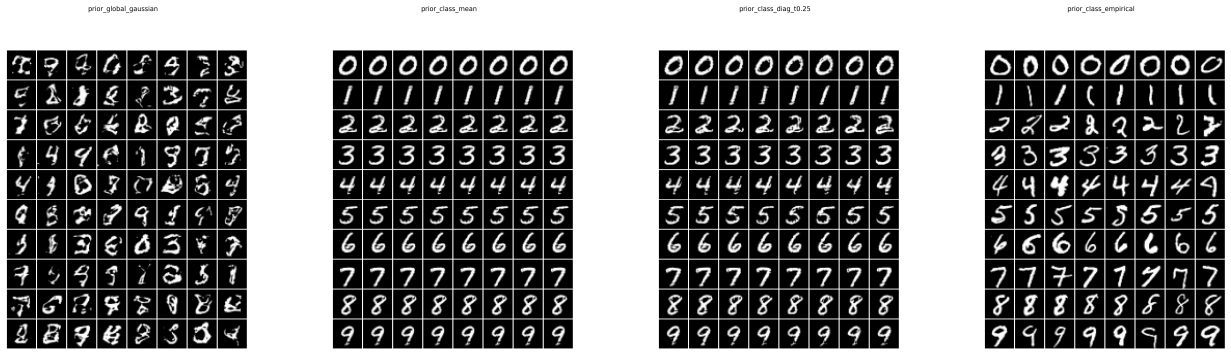


Figure 4: CIFAR-10 training curves. Phase transitions (epochs 50, 100, 200) are visible as step-changes; all terms converge smoothly.





(a) Global Gaussian
(self-ACC: 15%)

(b) Class mean style
(self-ACC: 100%)

(c) Class diag., $\tau=0.25$
(self-ACC: 100%)

(d) Empirical style bank
(self-ACC: 100%)

Figure 6: MNIST generation under representation-aware style sampling. Global Gaussian sampling produces shapeless images; all class-conditional strategies restore correct digit identity. Class diagonal Gaussian ($\tau = 0.25$) gives the best balance of class consistency and intra-class diversity.